

Shuping Li

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EDUCATION

- **Ph.D. in Global Hydrology** Tokyo, Japan
Department of Civil Engineering, The University of Tokyo 2020.09 – 2024.09
Advisor: Prof. Dai Yamazaki
- **M.E. in Hydraulic Engineering** Guangzhou, China
School of Geography and Planning, Sun Yat-sen University 2017.09 – 2019.07
Advisor: Prof. Guoping Tang
- **B.E. in Hydrology and Water Resources Engineering** Guangzhou, China
School of Geography and Planning, Sun Yat-sen University 2013.09 – 2017.07
Advisor: Prof. Guoping Tang

WORK EXPERIENCES

- **Project Researcher (Postdoctoral level)** Tokyo, Japan
Institute of Industrial Science, The University of Tokyo 2024.10 – Present
- **Research Assistant** Tokyo, Japan
Department of Civil Engineering, The University of Tokyo 2023.10 – 2024.09
- **Research Intern** Guangzhou, China
South China Institute of Environmental Sciences 2019.07 – 2020.08
- **Visiting Scholar** New Jersey, US
Civil and Environmental Engineering, Princeton University 2026.02
Department of Earth and Planetary Sciences, Rutgers University-New Brunswick 2023.03

RESEARCH INTERESTS

- Hydrological modeling at hillslope scale in land surface model
- Impact of hillslope water dynamics on land cover heterogeneity
- Interaction between soil moisture and vegetation dynamics
- Land-atmosphere interaction

PUBLICATIONS

M. First author and corresponding () author publications:*

- [M6] **Resolving Climate Heterogeneity along Hillslope in Land Surface Model**
Li, S.* et al.
In preparation
- [M5] **Kilometer-scale simulation systematically improves land–atmosphere coupling estimates**
Li, S.*, Tokuda, D., Hsu, H., Shih, C., Hsu, J., Lo, M., Takasuka, D., Tanoue, M., Yoshimura, K.
Under review
- [M4] **Resolving Land Cover Heterogeneity along Hillslope Improves Simulation of Terrestrial Water and Energy Budgets**
Li, S.*, Yamazaki, D., Tozawa, T., Adachi, K., Nitta, T., Zhao, G., Zhou, X., Yoshimura, K.
Water Resources Research, 61(9), 2025
- [M3] **Where in the World Are Vegetation Patterns Controlled by Hillslope Water Dynamics?**
Li, S.*, Yamazaki, D., Zhou, X., Zhao, G.
Water Resources Research, 60(4), 2024

- [M2] **Soil moisture-vegetation interaction from near-global in-situ soil moisture measurements**
Li, S.* and Sawada, Y.
Environmental Research Letters, 17(11), 2022
- [M1] **Concentration estimation of dissolved oxygen in Pearl River Basin using input variable selection and machine learning techniques**
 Li, W., Fang, H., Qin, G., Tan, X., Huang, Z., Zeng, F., Du, H.* and Li, S.*
Science of the Total Environment, 731, 139099, 2020

C. Other collaborative publications:

- [C2] **Developing a Levee Module for Global Flood Modelling with a Reach-Level Parameterization Approach**
 Zhao, G.*, Yamazaki, D., Tanaka, Y., Zhou, X., Li, S., Hu, Y., Hirabayashi, Y., Neal, J.C., Bates, P.D.
Water Resources Research, 61(8), 2025
- [C1] **Streamflow response to snow regime shift associated with climate variability in four mountain watersheds in the US Great Basin**
 Tang, G.*, Li, S., Yang, M., Xu, Z., Liu, Y., Gu, H.
Journal of Hydrology, 573(March), 255–266, 2019

BOOKS

- [B1] **Description of MATSIRO6**
 MATSIRO6 document writing team, Guo, Q., Kino, K., Li, S., Nitta, T., Takeshima, A., Suzuki, K. T., et al.
 Runoff (Chapter 9) and Tile scheme (Chapter 13), 2021

CONFERENCE PRESENTATIONS

O. Oral presentations:

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| [O10] | AOGS Annual meeting 2026
Revisiting Land-Atmosphere Coupling Across Spatial Scales: From Coarse to Kilometer-Scale Simulations | Fukuoka, Japan
2026.08 |
| [O9] | EGU General Assembly 2026
Revisiting Land-Atmosphere Coupling Across Spatial Scales: From Coarse to Kilometer-Scale Simulations | Vienna, Austria
2026.05 |
| [O8] | EALSA Workshop
Revisiting Land-Atmosphere Coupling Across Spatial Scales: From Coarse to Kilometer-Scale Simulations | Jeju, South Korea
2026.01 |
| [O7] | CAS-TWAS-WMO forum
Resolving Land Cover Heterogeneity Along Hillslope Improves Simulation of Terrestrial Water and Energy Budgets | Beijing, China
2025.10 |
| [O6] | AOGS Annual meeting 2025
Resolving Land Cover Heterogeneity Along Hillslope Improves Simulation of Terrestrial Water and Energy Budgets | Singapore
2025.07 |
| [O5] | JSHWR-JAHS conference 2024
Resolving land heterogeneity in land surface model modulates terrestrial water and energy budgets | Tokyo, Japan
2024.09 |
| [O4] | 9th GEWEX Conference
Representing hillslope-scale land surface heterogeneity in land surface model substantially modulates water and energy budget | Sapporo, Japan
2024.07 |
| [O3] | AGU Fall meeting 2022
Where can we find unique landscapes regulated by hillslope moisture dynamics? | Chicago, US
2022.12 |

[O2]	JSHWR-JAHS conference 2022 An efficient method to succinctly represent the explicit land cover heterogeneity formed by hillslope water dynamics	Kyoto, Japan 2022.09
[O1]	JpGU Annual meeting 2022 An efficient method to succinctly represent the explicit land cover heterogeneity formed by hillslope water dynamics	Chiba, Japan 2022.05
<i>P. Poster presentation:</i>		
[P5]	Land Surface Model Summit 2 Resolving Climate Heterogeneity along Hillslope in Land Surface Model	Rome, Italy 2026.09
[P4]	Hydro-JULES event Resolving Land Cover Heterogeneity Along Hillslope Improves Simulation of Terrestrial Water and Energy Budgets	London, UK 2025.09
[P3]	JpGU Annual meeting 2025 Resolving Land Cover Heterogeneity Along Hillslope Improves Simulation of Terrestrial Water and Energy Budgets	Chiba, Japan 2025.05
[P2]	AGU Fall meeting 2024 Resolving Hillslope Heterogeneities in Land Surface Model	Washington, D.C., US 2024.12
[P1]	AGU Fall meeting 2022 Soil moisture-vegetation correlation from 733 in-situ soil moisture observations	New Orleans, US 2021.12

INVITED TALKS

[T4]	National Institute of Natural Hazards Resolving Land Cover Heterogeneity along Hillslope Improves Simulation of Terrestrial Water and Energy Budgets	Beijing, China 2025.10
[T3]	IGSNRR, Chinese Academy of Science, Forum for Water Problem Resolving Land Cover Heterogeneity along Hillslope Improves Simulation of Terrestrial Water and Energy Budgets	Beijing, China 2025.10
[T2]	UKCEH-UTokyo, Global Hydrology seminar series Resolving Land Cover Heterogeneity along Hillslope Improves Simulation of Terrestrial Water and Energy Budgets	Online 2025.03
[T1]	Dept. of Hydraulic Engineering, Tsinghua University Soil moisture-vegetation correlation from 3239 in-situ soil moisture observations	Online 2022.01

GRANTS AND AWARDS

- **Kurata Grant** 2025.03 – 2026.03
The Hitachi Global Foundation (39 recipients out of 342 applicants)
- **Conference Travel support** 2024.07
9th GEWEX Open Science Conference
- **Musha Shugyo Travel Grant** 2023.03
School of Engineering, The University of Tokyo
- **Grant for Dispatch to International Research Meetings** 2022.12
Foundation for the Promotion of Industrial Science (FPIS)
- **Best Presentation in HSP2021 of Hydrosphere Environmental Group** 2021.07
Department of Civil Engineering, The University of Tokyo
- **MEXT scholarship** 2020.09 – 2023.09
Ministry of Education, Culture, Sports, Science and Technology of Japan

MEDIA REPORTS

- [R2] [Avoiding static land surface models: improvements in simulating water–energy–vegetation dynamics](#) 2025.09
Institute of Industrial Science, The University of Tokyo
- [R1] [Finding Where the Grass is Greener](#) 2024.05
Institute of Industrial Science, The University of Tokyo

TEACHING EXPERIENCE

- **Global hydrodynamics and its impact on ecosystem (guest lecture)** 2026.10
Tokyo City University

STUDENTS MENTORED

- **Siyuan Huang** UTokyo Master of Civil Engineering 2026.04 – 2026.08
- **Jules Barthes** University of Lyon, Visiting Master at UTokyo 2026.04 – 2026.09
- **Yuze Wang** UTokyo Master of GSFS → CUHK Ph.D. 2025.04 – Present
- **Shun Wakamatsu** UTokyo Master of Civil Engineering 2024.04 – 2026.03

ACADEMIC SERVICES

Reviewer for journal

- *Agroforestry Systems, CATENA, Environmental Research Letters, Earth System Dynamics, Frontiers in Plant Science, Journal of Advances in Modeling Earth Systems, Journal of Hydrology, Measurement Science and Technology, Scientific Reports.*

Convener for conference

- **EGU General Assembly 2026** Vienna, Austria
HS10.5/BG3: Ecohydrological responses to droughts in a changing environment: Mechanism, trends and impacts. 2026.05
- **6th Hydro90 meeting** Online
Mechanism and evolution of ecohydrological process 2026.05

Organizer for meeting and workshop

- **ILS model developer meeting** Chiba, Japan
2026.03
- **ILS model Couplethon workshop** Chiba, Japan
2026.03